

Fuzzy Feature Engineering for Urban Traffic Prediction: A Comparative Performance Analysis of LSTM, SNN, Random Forest, and LightGBM

Rishitha C, Divya Rithanya S, Karthikeya Y, Kaushal S, Manimaran S*

Amrita School of Artificial Intelligence,
Amrita Vishwa Vidyapeetham, Coimbatore, India
s_manimaran@cb.amrita.edu

Abstract

Urban traffic congestion is a severe challenges to mobility, productivity and emergency response in metropolitan areas. Traditional approaches such as road exapansion or fixed traffic signals are only short term relief and also they are lack of adaptability in dynamic conditions. In this work we proposes a Smart Traffic Management System using AI models with integration fuzzy logic to enhance human like prediction and interpretability in real time traffic control. We are used four different AI models they are Long Short term Memory(LSTM), Sequential Neural Network(SNN), Random Forest(RF), and Light Gradient Boosting(LightGBM). we implemented and compare using Bangalore Traffic Dataset.The system can categorize congestion into three easily understood levels: low, medium, and high, using fuzzy features based on membership functions and rule-based logic. This enhances performance and increases the interpretability of the outcomes.The fuzzy-logic-improved models all performed better than baseline ones in experiments: Regression accuracy improved significantly by LightGBM and Random Forest, whereas LSTM stood out especially in congestion classification. We created a web prototype named TrafficFlow AI to implement the work. The system assists functions such as optimized routes, traffic signal control, and even reduced CO2 output through real-time IoT information, interactive dashboards, and adaptive advisory support. Everything else being equal, this fuzzy-AI system presents an operable, scalable, and interpretable solution to traffic administration in smart cities.

Keywords: Data Preprocessing; Fuzzification; Rule Inference; Congestion Index; Decision Support System

1 Introduction

Traffic congestion in cities is among the most difficult issues in metropolitan cities. This has a direct effect on daily mobility and economic productivity as well as most seriously healthcare services, where medical emergency delays can prove to be life-threatening. Conventional solutions of road expansion, upgrading traffic control, and increasing public transport offer only short-term solutions [1, 2]. These solutions do not treat long-term increases in urban population and vehicle volume and frequently do not account for variables such as weather, highway conditions, and fluctuations in traffic flow [3].

To eliminate this challenge, we put forward Smart Traffic Management through Fuzzy Logic and AI Models. Smart Traffic Management renders conventional techniques adaptive through Fuzzy Logic by adapting signals, lanes, and paths in real time to ease congestion and make it efficient [4, 5].

Traffic congestion is brought about by many dependent relations and thus cannot be identified from a single relation alone. Conventional traffic control systems have no capacity to incorporate the nonlinear, uncertain, and interdependence of traffic in cities [6, 7]. Intelligent traffic control systems that have a capacity to use AI to predict traffic and make control decisions in real time to ease congestion are therefore required. Such models as LSTM, Neural Networks, Random Forest, and LightGBM have a capacity to learn traffic patterns and address high-order relations effectively [8, 9].

Traffic is neither completely clear nor jammed; it could be a mix of that. AI systems give precise outputs, but in real-life scenarios, we require output that is gradual. To further improve upon that, we also use Fuzzy Logic to impart human-like reasoning to articulate traffic congestion in terms of low, medium, or high instead of binary output [1, 5, 10].

Traditional traffic forecast models like ARIMA and Linear Regression have been employed extensively, but they're unable to handle the nonlinear nature and real-time complexity of modern traffic flow. [9]. To solve this, we use sophisticated AI and ML techniques. This paper compares two ML (Random Forest, LightGBM) and two deep learning (LSTM, Neural Networks) models and implements fuzzy logic to deliver human-like outputs to aid better decision-making [6, 8].

Aim of this work are to develop an intelligent traffic control system that incorporates AI models and fuzzy logic to improve traffic congestion classification and decision support. This work contributes a comparison of ML and deep learning models for traffic congestion forecasting and includes the development of TrafficFlow AI <https://kaushalrog.github.io/TrafficFlowAI/>, a prototype of a smart city website. This system allows the users to feed on traffic datasets, input real-time volume, speed parameters and output congestion index, signal time, travel time, and CO2 emissions. Fuzzy logic allows the traffic congestion to be categorized as low, medium, or high to simplify the system. For instance, if traffic density is high and average speed is slow, a system can extend green light to improve flow [5, 6, 11]. Through the incorporation of AI prediction and fuzzy logic, traffic control becomes efficient and able to reduce delays whilst improving mobility in cities. The main contributions of this paper are:

- Created a hybrid fuzzy AI model that combines fuzzy logic with LSTM, SNN, Random Forest, and LightGBM for urban traffic forecasting.
- Developed fuzzy features that represent the attributes of traffic as low, medium, and high which enhances the interpretability of the model and decision making.
- Created an all-inclusive comparison of some AI models on regression as well as classification problems using real traffic datasets.
- Developed TrafficFlow AI as an online, real-time data platform for the observation of traffic, indexing congestion, adaptive signal control, and CO2 estimation. an IoT data, predictive models, and fuzzy reasoning-based scalable and practical system for smart city traffic management

The rest of this paper is as follows. Section 2 summarizes literature about traffic prediction systems that is related. Section 3 explains our proposed system architecture and model in detail. Section 4 describes our comparative experiment results. Finally, Section 5 concludes this paper and also discusses our future work.

2 Literature Review

For short-term traffic forecasting, Liu et al. [1] proposed a fuzzy inference system from cloud model considering cloud-rule generation and fuzzy c-means clustering. While more fuzzy clusters allow higher computational complexity and the scalability issue of city-scale usage, it is noise-robust. Fuzzy feature modeling can be directly integrated into hybrid ML/DL pipelines.

In order to predict and visualize traffic, geographic information systems have been incorporated into fuzzy inference systems. To present maps that are interpretable and appropriate to planners of cities, Alkaissi [2] defined fuzzy rules to spatially attribute scores to congestion. While methods are most appropriate to offline analysis, they provide interesting design input to including fuzzy measurements of congestion and visualization into real-time dashboards.

Incomplete or lacking traffic data have also been dealt with by hybrid deep learning models. Pipelines that combine factor-aware LSTM models and data imputation considering exogenous variables such as weather and holidays were proposed by Zeng et al. [3]. While this approach makes prediction more robust, selecting appropriate external inputs becomes harder. By showing how edge sensor data, which is usually limited in practice, can be preprocessed before being combined with fuzzy and AI modules, it explains our design.

For traffic management, even hybrid fuzzy-AI methods have been explored. Fuzzy logic and deep Q-learning have been integrated by Tunc and Soylemez [4] to regulate traffic signals, achieving reduced waiting time and queue size along with improved RL convergence.

For long-term time series forecasting, Wang et al. [5] proposed a fuzzy inference-based LSTM (FLSTM). The approach augments accuracy and interpretability through fuzzy-rule forecasting combined with LSTMs to enhance memory. Nevertheless, limited experiments have been witnessed about this method in traffic investigation work, and difficulties of architectural complexity and sensitive tuning of fuzzy rule generation make it a challenging approach. This work presents a blueprint of integrating fuzzy congestion measures and LSTM, and it is just as similar in application to our hybrid traffic modeling framework.

Bidirectional GRU networks and certain deep networks have been used to forecast congestion classification. Soft-GRU models have been reported to offer lightweight substitutes to CNN- or transformer-based models by Abdullah et al. [6], and they can effectively capture congestion onset with good temporal modeling. Hybrid fuzzy-DL models have promise to enhance interpretability since fuzzy logic could not be incorporated and spatial modeling is low.

Goenawan [7] validated the potential of modular spatio-temporal pipelines by employing CNNs to capture spatio features and LSTM to give temporal outputs in self-controlling traffic systems, yet interpretability is restricted without fuzzy reasoning.

In real-time traffic control systems, a distributed multi-agent architecture that emphasized scalability has been proposed by Kadkhodayi et al. [8] but resulted in computational overhead and Internet of Things-based communication latency by utilizing fuzzy logic, neural networks, reinforcement learning, and Internet of Things.

Lin and Hsu [9] proposed fuzzy interval-based machine learning models to give interval predictions to Random Forest, LSTM, and BPN models so that uncertainty could be handled in ML predictions. While they are time-consuming and need to be designed carefully through membership functions, such interval-based predictions have uncertain decision-making.

In a review of adaptive fuzzy control and fuzzy-IoT integration, Gandrybida et al. [10] discovered strengths of uncertain information as well as demerits including implementation cost, parameter tuning challenges, and cybersecurity risk. Lastly, new work shows that when AI models apply fuzzy logic, traffic control systems interpretability and predictability improve gradually. Fuzzy logic can be used to assign a congestion level as low, medium, or high in an explanation that is consistent with human intuition, and time patterns that are nonlinear can be discerned by AI models. Scalability to facilitate mass deployment in cities is a compromise against real-time response and computational efficiency that is a problem. These findings are the impetus of this work that uses TrafficFlow AI in a fuzzy-AI hybrid system to combine understandable decision-making and predictive modeling to enhance traffic flow in cities.

3 Proposed Work

First, from a research perspective, we discuss developing a fuzzy-AI hybrid modeling framework designed to accurately predict traffic congestion despite inherent uncertainties in the data. The second contribution, which is the applicability aspect at the city-level use, takes these models and builds it into a web-based Traffic Management Control System (TrafficFlow AI) for use by traffic police and other city-level administrators. Collectively, the contributions bridge together a research scientific rigor and practical use of smart city traffic management as shown in Figure 1.

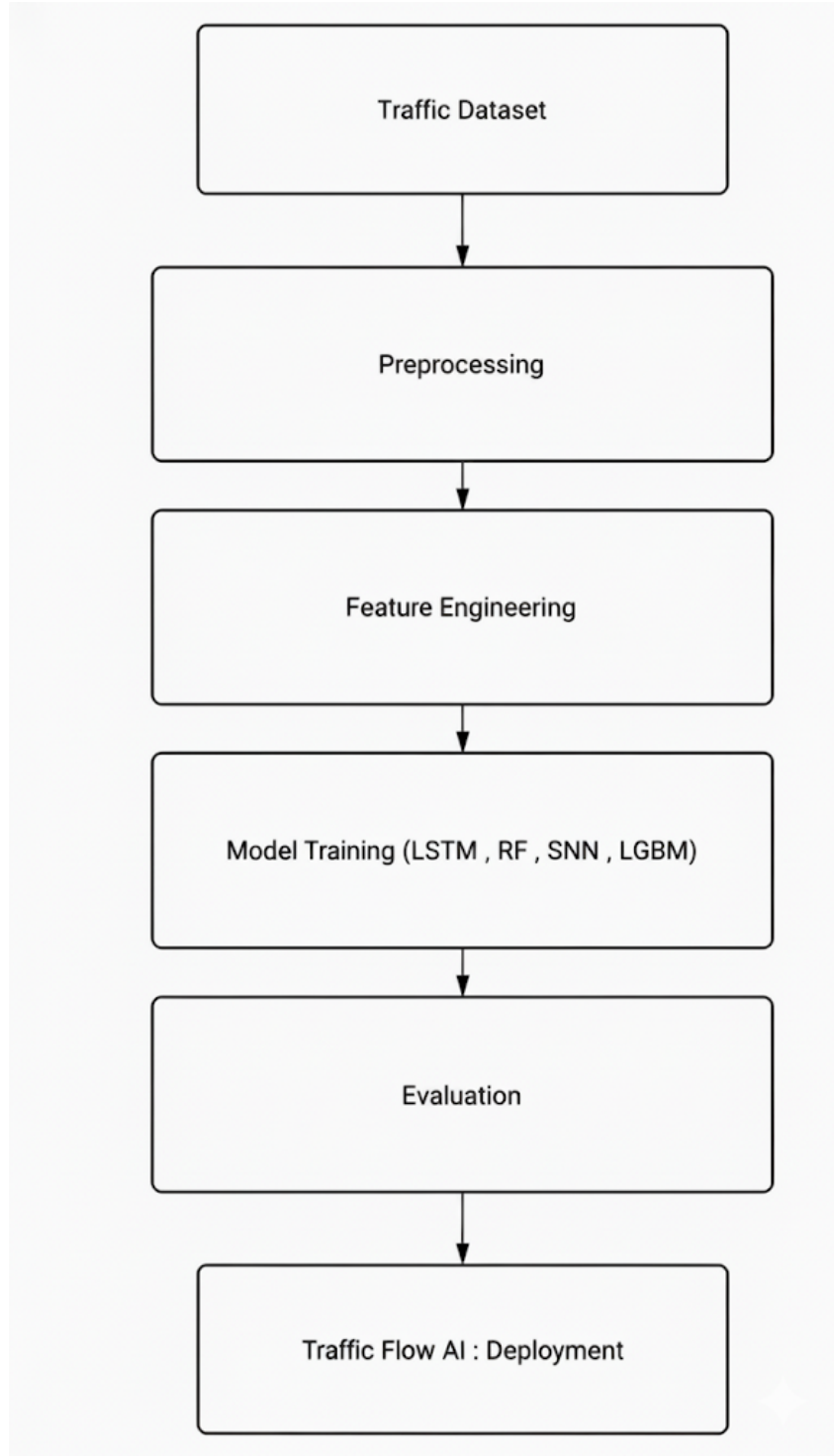


Figure 1: System Architecture

3.1 Proposed Work I: Fuzzy–AI Modeling for Traffic Problem Prediction

This part addresses about the scientific research about developing a model using fuzzy logic for improving prediction accuracy in user point of view.

Architecture & Workflow

- **Phase 1: Data Preprocessing** – The Bangalore Dataset [12] contains (8,936 entries, 16 features)

is used for this research. Data Preprocessing includes handling missing values, normalization and encoding[3, 13].

- **Phase 2: Fuzzification** – Non linear values such as traffic volume, speed, and congestion level are classified into Low, Medium, High categories. Triangular membership functions are deployed[1, 2, 11].
- **Phase 3: Rule Inference** – Fuzzy IF-THEN rules ensure transparent reasoning for congestion. Example: IF Volume = High AND Speed = Low THEN Congestion = High[4, 10].
- **Phase 4: Model Training and Evaluation** – Four models (LSTM, SNN, RF, LGBM) are trained in baseline (raw features) vs fuzzy-enhanced (raw + fuzzy). Evaluated with regression (MSE, RMSE, MAE, R^2) and classification (Accuracy, Precision, Recall, F1)[5, 9, 14].
- **Phase 5: Results Integration** – The hybrid fuzzy-AI models produced superior results compared with the baselines, validating fuzzy feature engineering[6, 7].

3.1.1 Model Implementation

Detailed implementation of LSTM, SNN, RF, LGBM for traffic forecasting with and without fuzzy features is presented (including preprocessing, architectures, evaluation).

1. Long Short-Term Memory (LSTM)

Data preparation phase filled missing data through forward filling, and all features were normalized through a `MinMaxScaler` to allow for constant scaling[3, 13]. To preserve temporal dependencies, the data were transformed through a sliding window of 10 time steps to create sequences[6, 14]. The dataset was then divided chronologically into training (80%), and testing (20%) sets[12]. In the baseline method (no fuzzy features), two different tasks were evaluated for the model. For **regression**, it included an LSTM network (64 units, ReLU activation) followed by a dropout layer (rate of 0.3) and a single-neuron dense output layer[5].

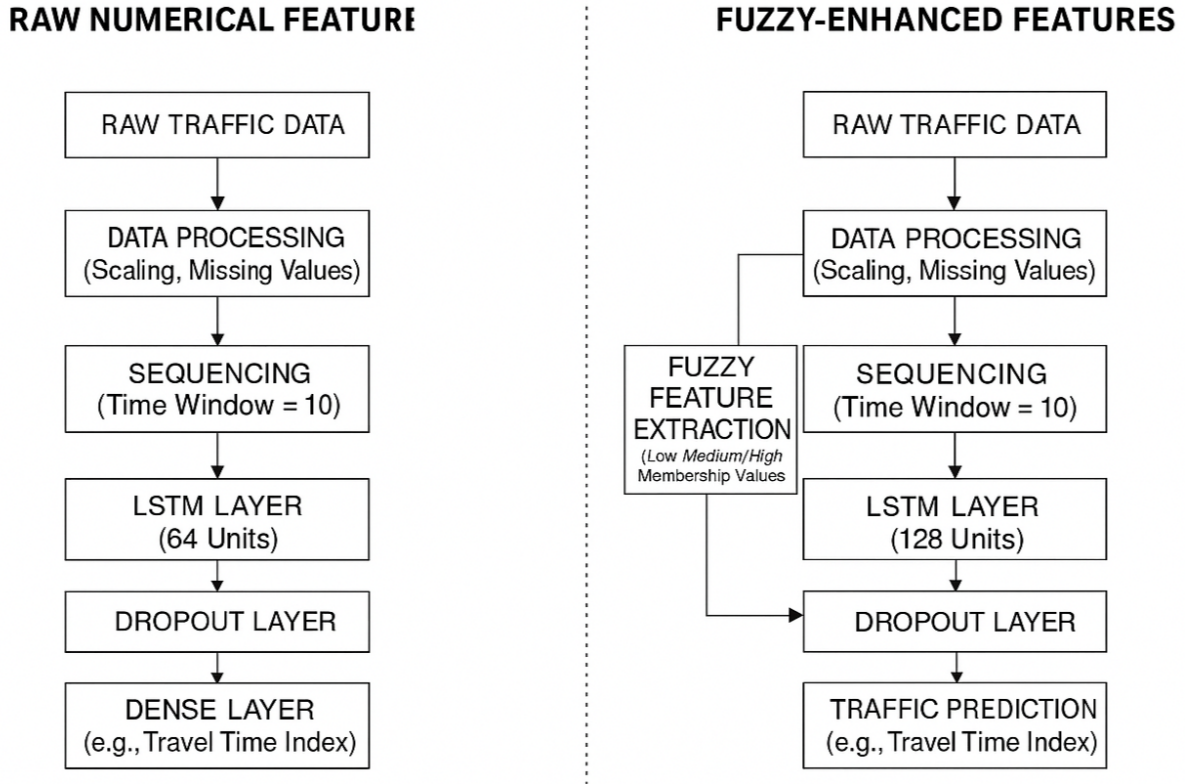


Figure 2: LSTM Workflow: A comparison between the model using raw numerical features and the model enhanced with fuzzy features

For **classification**, a stacked LSTM structure (128 and 64 units) was used combined along with dropout layers and ended up with a dense softmax output layer to predict category as depicted in Figure 2. These two models were optimized with the Adam optimizer while mean squared error (MSE) is minimized to achieve regression and sparse categorical cross-entropy to achieve classification.

Fuzzy-enhanced method supplemented input data through adding membership values from three fuzzy functions (Low, Medium, and High) of important indicators. Regression tasks of performance appraisal were evaluated through MSE, RMSE, MAE, and R^2 [13], and classification accuracy was evaluated by accuracy, precision, recall, F1-score, and the confusion matrix. **Insight:** The key finding is that the fuzzy-enhanced LSTM demonstrated superior performance in handling uncertainty and offered greater model interpretability compared to the standard LSTM.

2. Sequential Neural Network (SNN)

Data preparation consisted of eliminating non-numeric variables, gap filling through forward filling, and normalizing variables through `MinMaxScaler`[3, 13]. Categorical labels were expressed numerically and the data set enriched by including fuzzy membership degrees as extra variables to distinguish qualitative patterns [1, 2].

The baseline model employed a Sequential Neural Network (SNN) structure consisting of Dense layers (128, 64, and 32 units with ReLU activation function), which were separated by dropout layers to prevent overfitting. For **regression** tasks, a single linear output layer was used, while for **classification**, a softmax output layer served multi-class outputs as depicted in Figure 3. Models were optimized against the Adam optimizer, mean squared error (MSE) for regression and sparse categorical cross-entropy loss for classification tasks, and generally ran for 20–50 epochs[6, 7, 14].

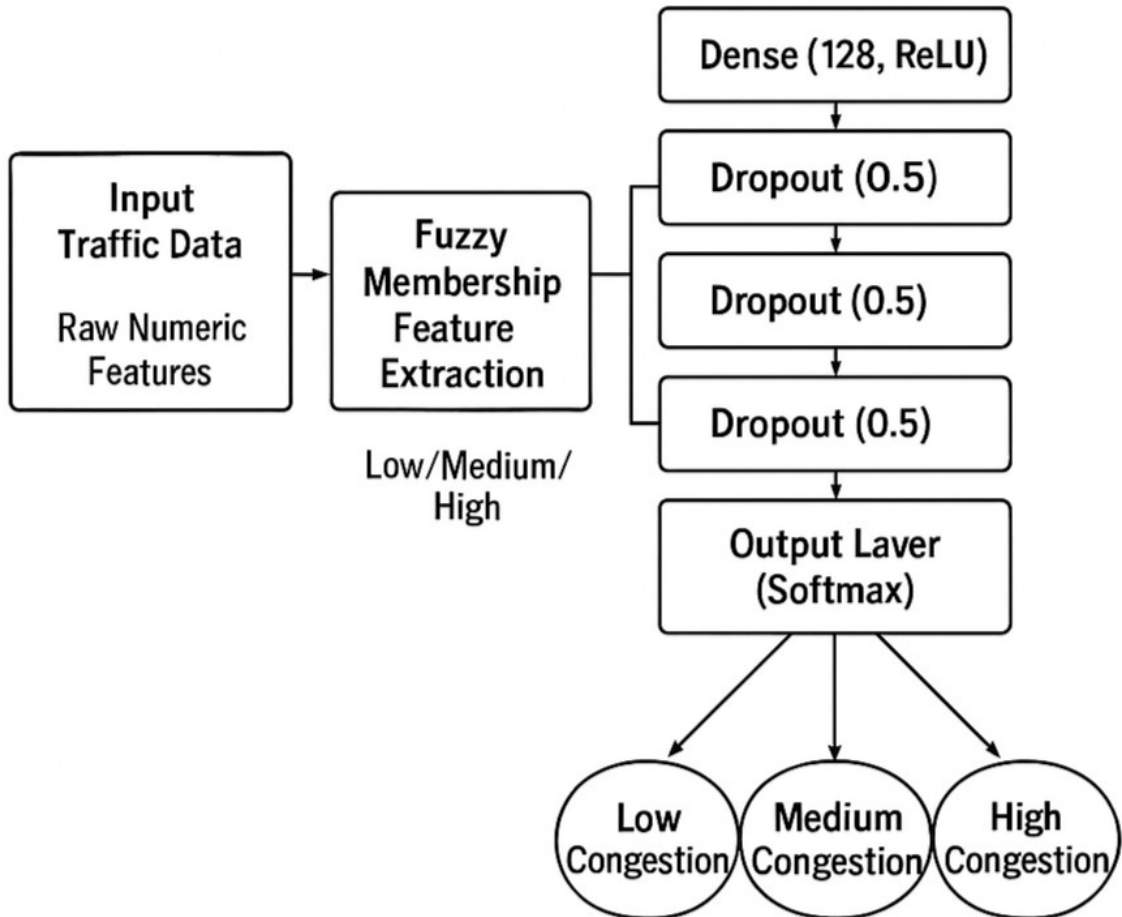


Figure 3: SNN classification workflow with and without fuzzy features.

Regarding the fuzzy-enhanced model, the three membership degrees of Low, Medium, and High status were attached to salient input features (ex. Travel Time Index and Traffic Volume) [1, 11]. The relative features of this expansion were then given to another exact copy of the SNN architecture. The performance of the models was evaluated with typical metrics: regression [MSE, RMSE, MAE, R^2], and classification [accuracy, confusion matrix, classification report] [4, 11].

3. Random Forest (RF)

For data preparation, categorical variables were omitted and missing values treated by forward filling. To ensure uniformity in the analysis, all numerical attributes were normalized. Subsequently, we broadened the feature set by incorporating fuzzy membership values calculated from the Travel Time Index and Traffic Volume, as a means of addressing the uncertainty and fuzziness involved in traffic data [1, 2, 10] as depicted in Figure 4.

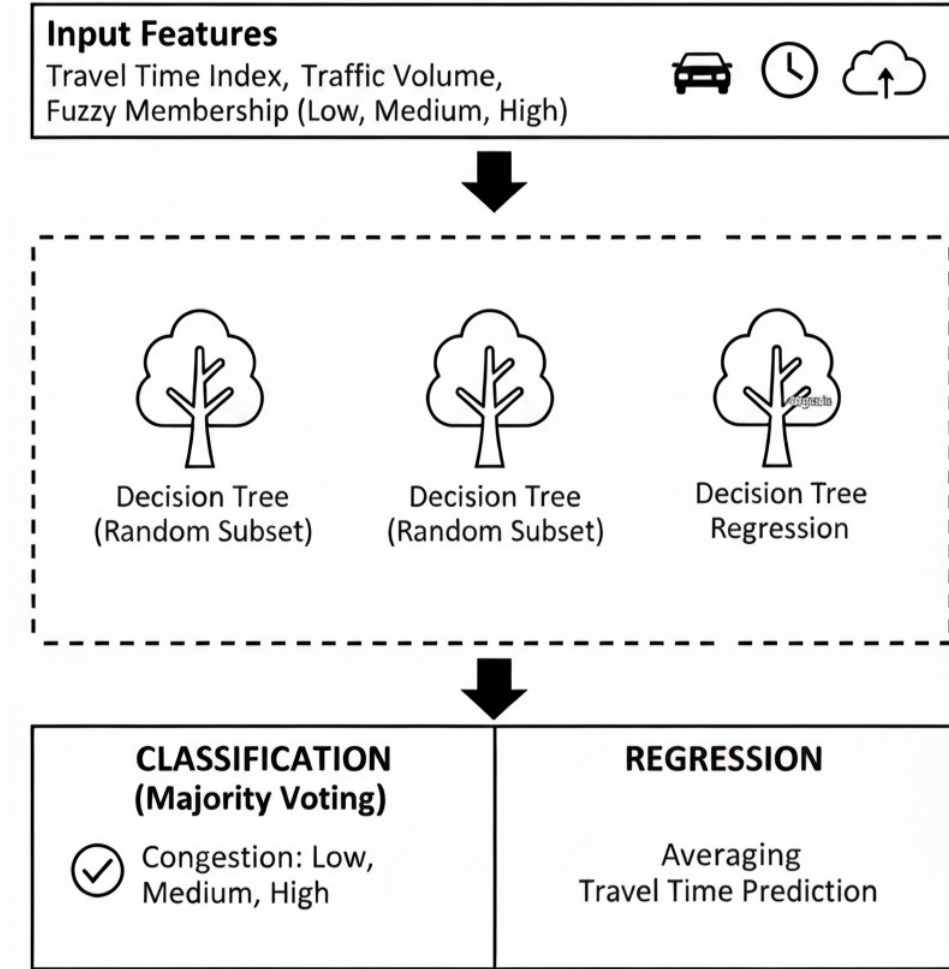


Figure 4: Random Forest workflow for traffic prediction.

The `RandomForestRegressor` (for regression) and a `RandomForestClassifier` (for classification) as baseline model (without fuzzy features) was built with a of 100 decision trees for regression, and a of 100 decision trees for classification. With a re-fuzzy model created by adding membership values, we allowed for greater capture of the unexplained uncertain congestion states through decision trees [11]. To take advantage of the superior features at the higher levels, we increased our model capacity of trees to 200 trees, to introduce a more robust model of revealing complex traffic patterns based on traffic data [8].

Model performance was evaluated using common metrics. For regression, the metrics were mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2) [13]. For classification, accuracy, precision, recall, F1-score were used along with a confusion matrix to give a thorough analysis [4, 11].

4. LightGBM

Pre-processing chain began by removing irrelevant columns to clean up the dataset. Numerical variables were normalized and categorical variables were label-encoded to make it suitable to fit into models. Fuzzy membership variables were included in suitable numerical columns as well to enhance feature representation[1, 2, 10].

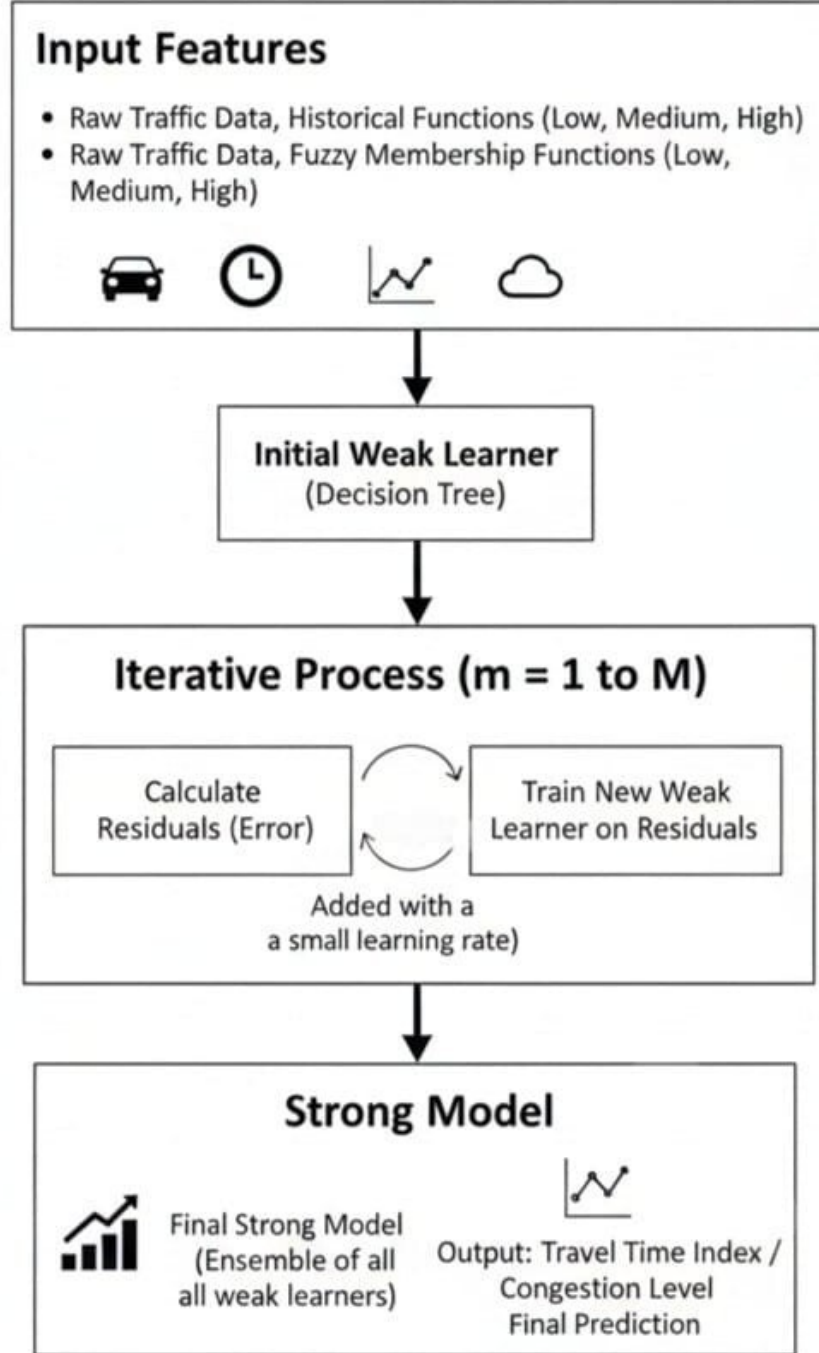


Figure 5: : LightGBM workflow for regression and classification tasks.

The model used a LGBMRegressor to run the regression tasks, with 500 estimators and 0.05 learning rate. For classification tasks, a LGBMClassifier ran with default parameters[6, 7]. For the fuzzy-enhanced method, the model was re-optimized to make use of the new expanded set of features; hyper parameters were optimized to 100 estimators, 10=max depths, and 1.0 learning

rate[8] as shown in Figure 5.

Standard metrics were implemented to evaluate performance. In the case of regression problems these metrics included: mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and R^2 [13]. For classification problems, accuracy was done through the analysis of a confusion matrix, along with a complete classification report to determine predictive power overall, as well as performance based on class.

3.1.2 Comparative Observations

- **Baseline (Non-Fuzzy) Models:** Built from raw numerical features (traffic volume, travel time index). *Advantages:* High predictive accuracy in regression tasks. *Disadvantages:* Low interpretability, harder to relate predictions to intuitive traffic categories. Example: LSTM captured temporal dependencies but was difficult to interpret.
- **Fuzzy-Enhanced Models:** Incorporated semantic categories (Low, Medium, High) via fuzzy membership functions. *Advantages:* Better interpretability and improved accuracy. *Impact:* RF and LightGBM showed notable accuracy gains; LSTM gained human-like reasoning ability alongside time-series modeling.

3.2 Proposed Work II: TrafficFlow AI Web Platform for City Traffic Management Control

3.2.1 Objective

The primary objective of the TrafficFlow AI web platform is to provide assistance to city governments and traffic police officers in managing or controlling traffic flow through a decision support tool using AI. Navigation apps may be available to the public but this web and AI platform is designed to be used in municipal traffic control centers to maximize urban, sustainable, efficient, and safe traffic patterns.

3.2.2 System Architecture

The overall system architecture (Figure 6) is comprised of three functional layers:

- **Data Input Layer**
 - Accepts uploaded traffic datasets.
 - Connects to real-time IoT traffic sensors and surveillance streams.
 - Supports manual entry of context-dependent parameters (e.g., accidents, weather).
- **AI Engine Layer**
 - Fuzzification of imprecise traffic parameters such as volume, speed, and density.
 - Rule-based inference for congestion hotspot identification.
 - Hybrid AI model execution including:
 - * Long Short-Term Memory (LSTM),
 - * Sequential Neural Networks (SNN),
 - * Random Forest (RF),
 - * Light Gradient Boosting Machine (LGBM).
- **Decision and Control Layer**
 - Visual interactive dashboards.
 - Automated incident notifications.
 - Optimization models for adaptive traffic management.

3.2.3 Phases of Operation

The TrafficFlow AI program methodology follows structured steps similar to city traffic management practices as shown the Figure 6:

1. **Data Gathering:** Batch entry of datasets by operators or direct IoT sensor feeds across the city.
2. **Configuration:** Authorities simulate varying scenarios (e.g., accidents, rush hour, crowd gatherings) for situational awareness.
3. **AI Analysis:** Hybrid models predict congestion levels, travel and waiting time, and environmental impacts (e.g., weather).
4. **Real-time Monitoring:** Dashboards display congestion percentages, flow rates, and incident reports dynamically.
5. **Optimization Control:** Provides prescriptive measures such as:
 - Adaptive green-signal allocation,
 - Diversion to side streets,
 - Eco-prioritized scheduling for reducing delays and emissions.
6. **Closed Feedback Loop:** The system adapts recommendations using new data, shifting from static planning to real-time adaptive traffic management.

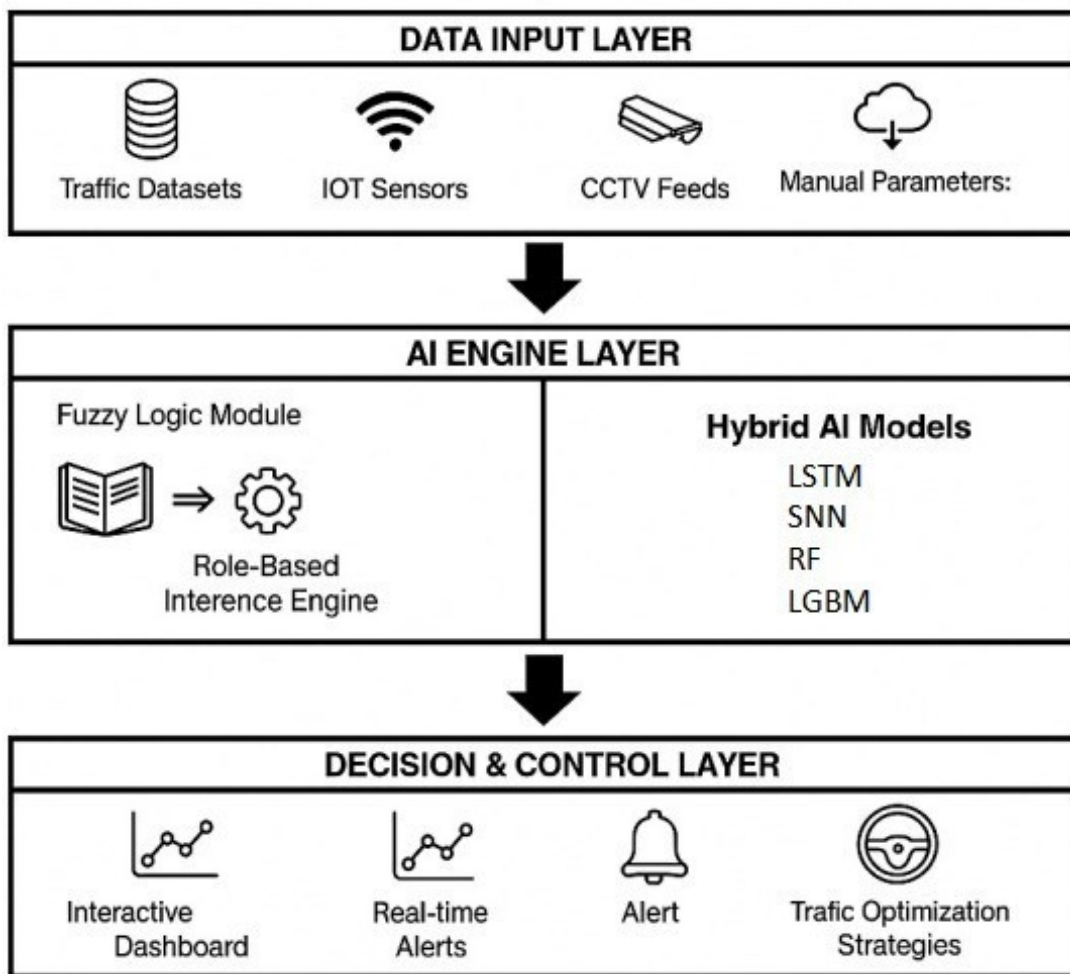


Figure 6: Overall System Architecture

3.2.4 Algorithmic Workflow

TrafficFlow AI Control System (Algorithm 2) itself is the core of the platform. This is quite distinct from time-rule or fixed-rule systems and is itself a closed-loop adaptive system that needs constant real-time feedback. It incorporates three distinguishable paradigms: fuzzification, hybrid AI reasoning and dynamical control optimisation [7, 8].

The process begins with data import and preprocessing. Traffic data is collected from diverse sources, including police records, sensors, and operator input. These feeds are then standardized, cleaned, and rescaled to maintain information coherence. Crucially, real-time variations like traffic waves are preserved to ensure the system retains situational realism. [3, 13].

Noise robustness is offered by Random Forest (RF) models in ensemble learning[8]. Outputs from these models are transformed to actionable information through interactive dashboards, congestion indices updating in real time, wait time, emissions, and travel time duration, while anomaly alerts are transmitted instantly to operators to allow timely intervention [7]. Optimisation and recommendation of control integrate adaptive intervals of signals, alternate path methods, and environmentally-optimised cycles to reduce emissions [4, 11].

Finally, the system integrates continuous learning such as new information regarding traffic, recalculation of congestion threshold dynamically, and retraining lightweight models when significant deviations are identified [1, 5].

Section 4 presents experiment results of depicting the quantitative superiority of fuzzy attributes by LSTM, SNN, RF, and LGBM in regression and classification tasks. We further illustrate the contribution of TrafficFlow AI engine in providing real-time decision support by traffic authorities.

4 Results and Discussion

4.1 Experimental Results

In this sub-section, we will evaluate the performance of each of the four models—LSTM, SNN, Random Forest, and LightGBM—on two regression tasks (Travel Time Index, Traffic Volume) and a classification task (Congestion Level) [3]. Each model was evaluated both in a baseline setting (numerical features only) and in a fuzzy setting (numerical + fuzzy membership values). Therefore, the next couple of bullet-point descriptions will detail the results in terms of accuracy, tradeoffs, and implication for use in traffic management [8].

4.1.1 Regression Results: Travel Time Index Prediction

Table 1 shows the results for Travel Time Index (TTI), which directly measures congestion level and affects travel-time [2].

The tree-based ensemble methods like **Random Forest** and **LightGBM** showed **significant improvement** when using the fuzzy features. They reached near-perfect prediction accuracy ($R^2 \approx 0.999$), compared to baseline R^2 values of 0.800 and 0.773. The RMSE values also decreased by over 99%. The fuzzy membership values effectively captured the nonlinear uncertainty of traffic, which the ensemble methods leveraged well [1].

In contrast, the performance of LSTM and SNN decreased on fuzzification, with SNN decreasing significantly from $R^2 = 0.711$ to negative, and only marginal degrade was seen with LSTM. This suggests that time deep models struggle to utilize fuzzy categorical semantics for continuous prediction tasks [5, 9].

4.1.2 Regression Results: Traffic Volume Prediction

Table 2 presents the regression results for the estimation of traffic volume. Predicting the correct flow aids lane assignment as well as congestion control [13].

Table 1: Regression Results for Travel Time Index Prediction

Model	Features Used	MSE	RMSE	MAE	R^2
LSTM	Baseline	0.027	0.165	0.149	-0.022
	Fuzzy-Enhanced	0.027	0.164	0.143	-0.043
SNN	Baseline	0.008	0.090	0.057	0.711
	Fuzzy-Enhanced	0.027	0.164	0.146	-0.010
Random Forest	Baseline	0.005	0.074	0.040	0.800
	Fuzzy-Enhanced	5.47e-09	7.40e-05	3.52e-05	0.999
LightGBM	Baseline	0.006	0.070	0.044	0.773
	Fuzzy-Enhanced	4.27e-08	0.0002	0.0001	0.999

LSTM and SNN decreased significantly after fuzzification. For instance, SNN decreased from $R^2 = 0.858$ to almost zero. It supports the fact further that the semantics of fuzziness blend well when used with tree-based learners as compared to deep time networks [14].

Table 2: Regression Results for Traffic Volume Prediction

Model	Features Used	MSE	RMSE	MAE	R^2
LSTM	Baseline	1.67e+08	12931.3	10607.3	-0.013
	Fuzzy-Enhanced	1.72e+08	13113.6	10902.8	-0.042
SNN	Baseline	2.44e+07	4939.9	3831.2	0.858
	Fuzzy-Enhanced	1.66e+08	12869.0	10482.0	-0.004
Random Forest	Baseline	819.94	28.63	7.09	0.999
	Fuzzy-Enhanced	123.27	11.10	5.07	0.999
LightGBM	Baseline	42528.8	206.23	73.79	0.999
	Fuzzy-Enhanced	2042.96	45.20	31.04	0.999

4.1.3 Classification Results: Congestion Level

Table 3 This summarizes the performance of the model at classifying levels of congestion [6]. In this task, it was the deep models that benefited the most; LSTM improved accuracy from 0.73 to 0.88 and the F1-score from 0.62 to 0.87 [5]. LightGBM also improved considerably - 0.64 to 0.91 - after fuzzification. In general, these support the value of fuzzy semantics in providing and augmenting weaker classifiers [10].

The SNN and RF had already achieved high baseline accuracy (~ 0.94) so there were only small steps forward (0.95) in accuracy. Stability equates to the network generalizing well across the traffic states.

4.1.4 ML vs. DL Comparison

To synthesize the findings, Table 4 contrasts machine learning (ML) and deep learning (DL) performance across regression and classification.

Table 3: Classification Results for Congestion Level

Model	Features Used	Accuracy	Precision	Recall	F1-score
LSTM	Baseline	0.73	0.54	0.73	0.62
	Fuzzy-Enhanced	0.88	0.91	0.88	0.87
SNN	Baseline	0.94	0.88	0.89	0.88
	Fuzzy-Enhanced	0.95	0.95	0.95	0.95
Random Forest	Baseline	0.94	0.94	0.94	0.94
	Fuzzy-Enhanced	0.95	0.95	0.95	0.95
LightGBM	Baseline	0.64	0.63	0.64	0.61
	Fuzzy-Enhanced	0.91	0.90	0.91	0.91

4.1.5 Comparative Analysis

The comparative analysis reveals a division of strengths:

- **Regression tasks** were so dominated by ensemble ML methods, Random Forest the most reliable and LightGBM is also effective [13].
- **Classification tasks** were done by deep learning models, especially LSTM, which benefited from fuzzification [7].
- **Fuzzy features** reshaped the performance in some task-specific ways—boosting ML ensembles in regression and DL models for classification [1, 5].

In terms of applications, these results suggest that hybrid deployments are preferable [14]. ML ensembles will be ideal for continuous traffic forecasting and DL models will be well-suited for categorical predicting congestion. This complementary specialization will ensure prediction is both accurate as well as interpretable and actionable for practical traffic applications [8].

Table 4: Improvement Summary for Travel Time Index (Baseline → Fuzzy)

Model	Baseline R^2	Fuzzy R^2	Change
LSTM	-0.021	-0.043	↓ 0.022
SNN	0.711	-0.010	↓ 0.721
Random Forest	0.800	0.999	↑ 0.199
LightGBM	0.773	0.999	↑ 0.226

4.2 Website Results

While section 4.1 demonstrated how the AI models could perform in isolation, this section presents how they could be incorporated into a web-based decision-support system [7]. A fictitious dataset similar to real urban traffic was performed on the website. The testing was performed for two reasons: (i) to technically test the website for processing the data flows in real time, and (ii) to practically test the interface for actual traffic management [8].

4.2.1 From Input to Analysis

Figure 7 demonstrates the configuration module, which is responsible for users identifying key parameters, for example, traffic volume, density, speed, population density, time of day, and weather [3]. Once you get

to this stage you are able to be flexible with the simulation of various scenarios, for example, simulating peak periods during the evening rush hour, or simulating low density off-peak periods. When you finish configuring the settings, the system automatically calls AI models to produce predictive outputs.

Figure 7: Traffic configuration interface for setting simulation parameters.

4.2.2 Comprehensive Traffic Insights

As shown in Figure 8, the analysis dashboard consolidates outputs from regression and classification tasks. Beyond raw congestion indices, it translates predictions into operational indicators such as travel time, traffic efficiency, CO₂ emissions, and signal timing recommendations [2].

Key Insights:

- Regression results (e.g., 18.6 min travel time) are directly linked to actionable policies such as route adjustments.
- Classification of congestion into levels (Low/Medium/High) makes results interpretable at a glance.
- Environmental and efficiency metrics (e.g., 168g CO₂/km, 75% efficiency) provide a holistic view beyond mobility.



Figure 8: Traffic analysis results summarizing congestion index, efficiency, and environmental impact.

4.2.3 Live Monitoring Capabilities

The platform also supports live monitoring (Figure 9). Real-time dashboards track thousands of vehicles simultaneously, with automatic incident detection and efficiency tracking [8]. A time-series visualization shows how traffic volume, average speed, and system efficiency evolve during the day.

Performance Observations:

- Scalable processing: over 2,700 vehicles tracked in parallel.
- Fast predictions: more than 140 AI inferences per minute with sub-2s latency.
- Reliable reporting: system efficiency consistently around 90%.



Figure 9: Live monitoring dashboard with vehicle counts, incidents, and performance metrics.

4.2.4 Optimization for Decision-Making

Perhaps the most practical feature is the optimization module (Figure 10), which transforms predictions into control actions. Using adaptive signal control, the system automatically suggests modifications in green signal time and prioritization algorithms [4, 11].

Results:

- Traffic flow improved by +36% compared to baseline conditions.
- Delay reduced by -32% and emissions cut by -20%.
- AI accuracy of predictions within this module reached 98.3%.

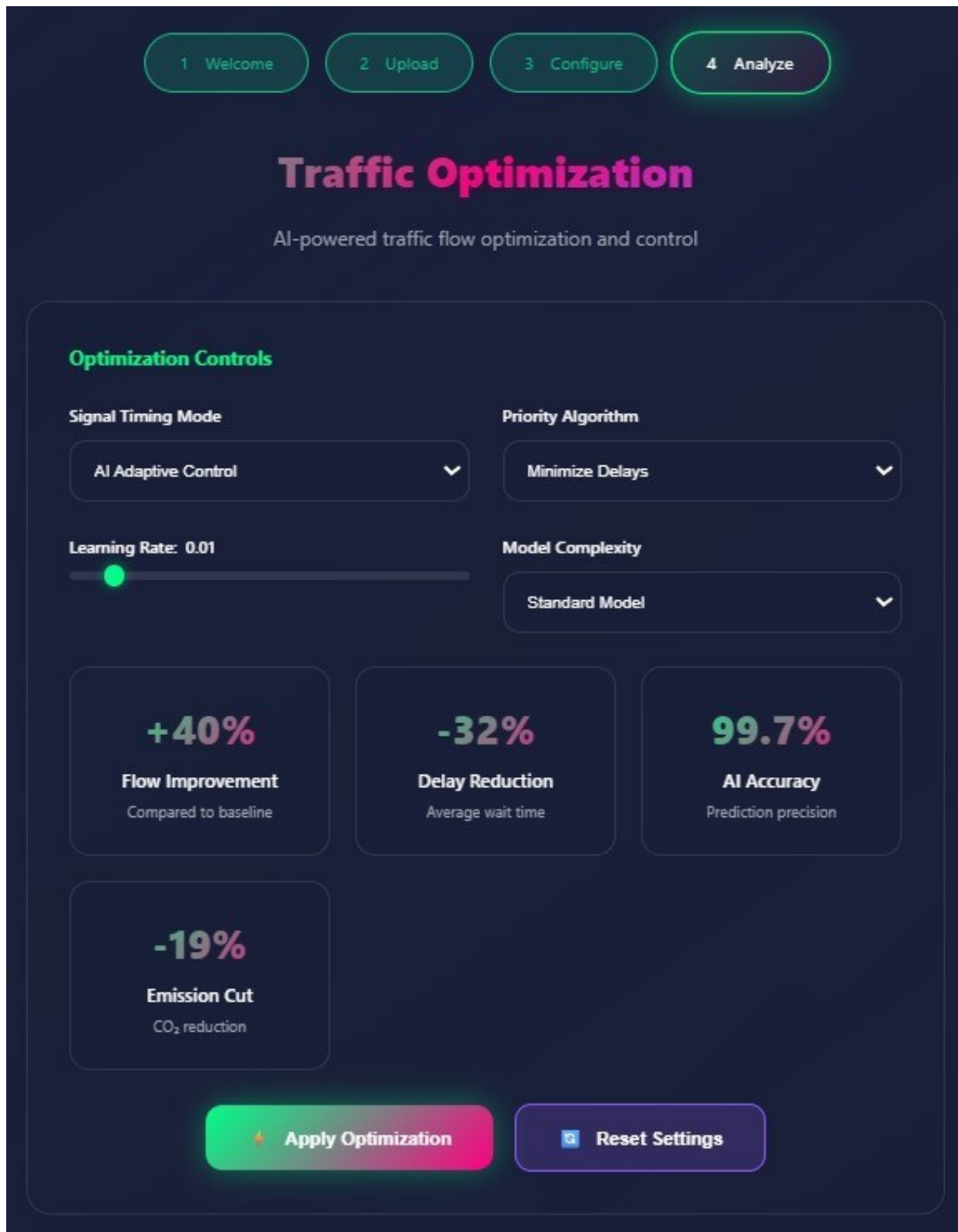


Figure 10: Optimization interface showing adaptive control and resulting performance gains.

4.2.5 Integrated Contribution of Modules

To highlight the synergy of the system, Table 5 summarizes how each module contributes to actionable traffic control. Unlike experimental results in isolation, the website demonstrates how regression and classification outputs are operationalized into monitoring and optimization for real-world scenarios.

4.2.6 Overall Implication

The website results establish that the AI models, when embedded into an interactive platform, move beyond predictive accuracy to actionable intelligence. The platform combines fast response times (sub-2s

Table 5: Integrated Contribution of System Modules to Traffic Management

Module	Outputs Provided	Contribution to Traffic Management
Regression	Travel Time Index, Traffic Volume	Provides quantitative forecasts that guide route optimization, congestion alerts, and proactive planning.
Classification	Congestion Level (Low/Medium/High)	Enables rapid interpretability for non-technical operators, assisting in adaptive signal timing decisions.
Monitoring	Vehicle counts, incidents, system efficiency	Offers real-time visibility into the state of the network, enabling timely detection of bottlenecks or disruptions.
Optimization	Signal timing, flow improvements, emission reduction	Converts predictions into direct operational controls that reduce delays, improve flow, and enhance sustainability.

latency), scalable computation (140+ predictions per minute), and clear interpretability (simple congestion levels and signal recommendations). This combination ensures that even non-expert operators can rely on the system for traffic control. Although evaluated on synthetic data, the platform demonstrates proof-of-concept feasibility and positions itself as a deployable solution once live traffic feeds are integrated. To improve this website in future by using cloud data management and using highend models like CNN and Vision transformers to increase the accuracy and to provide better results for avoiding such traffic situations in the coming days.

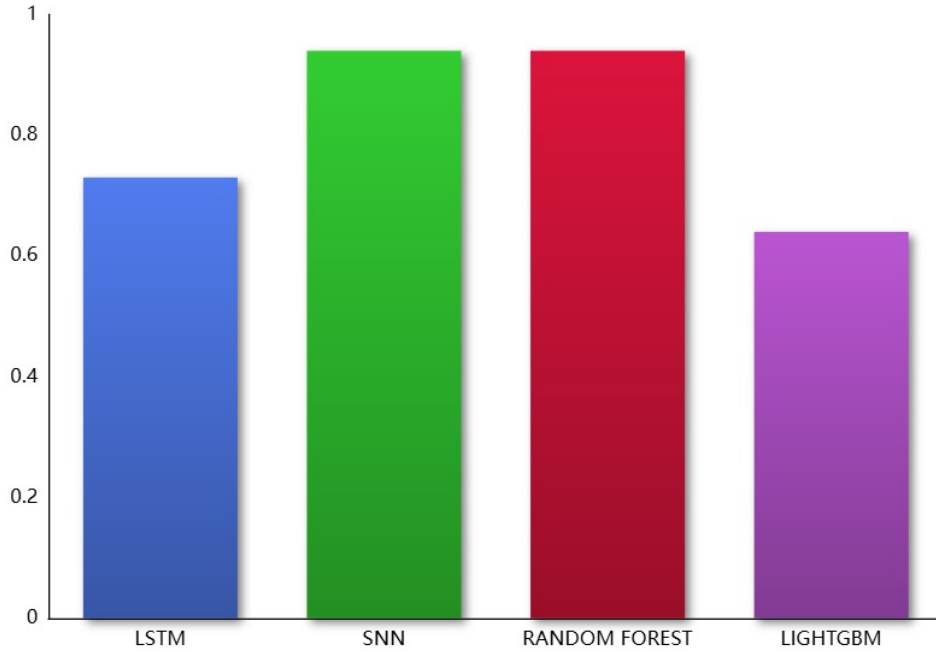


Figure 11: Baseline accuracy.

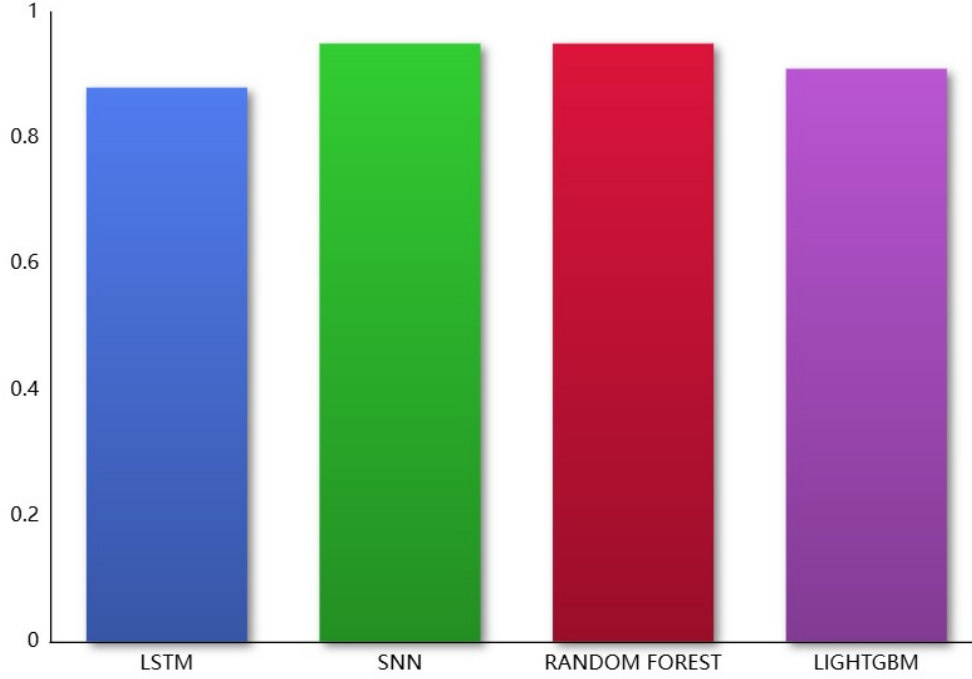


Figure 12: Fuzzy enhanced accuracy.

5 CONCLUSION

In this work, we tried to solve one of the biggest issues in cities today which is Traffic Congestion. The idea of using Artificial Intelligence (AI) models along with fuzzy logic to make traffic management smarter and more adaptive is what we have implemented here. We compared two Machine Learning models (Random Forest, LightGBM) and two Deep Learning models (Long Short-Term Memory - LSTM, Sequential Neural Networks - SNN) for their performance in predicting the outputs. From our experiments, LSTM performed the best in terms of prediction accuracy. By adding fuzzy logic, the system could translate raw predictions into clear levels like low, medium, and high congestion, which makes the results easier to understand and apply in real-time. We also built a web-based prototype, TrafficFlow AI, that provides live traffic monitoring, congestion index, optimized green signal timing, and even estimates the travel time and CO2 emissions.

Future improvements could include integrating Reinforcement Learning for adaptive controls, and extending this system to cover larger areas with additional features that make it even smarter.

COMPETING INTERESTS

The authors declare that they have no competing interests.

FUNDING INFORMATION

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

AUTHOR CONTRIBUTION

Rishitha C, Divya Rithanya S, Karthikeya Y, and Kaushal S contributed equally to this work. We were responsible for data collection and preprocessing, feature engineering, design and implementation of the fuzzy logic framework, development and training of machine learning and deep learning models, experimental evaluation, and analysis of results. The four authors also jointly prepared the initial draft of the manuscript and revised it based on feedback.

Manimaran S served as the corresponding author and supervised the entire research process. He provided conceptual guidance, defined the research direction, reviewed the methodology and experimental design, and critically revised the manuscript for technical accuracy and clarity.

DATA AVAILABILITY STATEMENT

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

RESEARCH INVOLVING HUMAN AND/OR ANIMALS

Not Applicable.

INFORMED CONSENT

Not Applicable.

ETHICAL STATEMENT

This research does not involve any human subjects or animals and uses only publicly available traffic datasets. No personal data was collected. The proposed system functions as a decision-support tool and it is designed in accordance with ethical principles of transparency, fairness, and responsible AI deployment.

References

- [1] H. Liu, Y. Wang, X. Wang, Y. Liu, Y. Liu, X. Zhang, F. Xiao, Cloud model-based fuzzy inference system for short-term traffic flow prediction, *Mathematics* 11 (11) (2023) 2509. [doi:10.3390/math11112509](https://doi.org/10.3390/math11112509).
- [2] Z. A. Alkaissi, Traffic congestion evaluation of urban streets based on fuzzy inference system and gis application, *Ain Shams Engineering Journal* 15 (6) (2024) 102725. [doi:10.1016/j.asej.2024.102725](https://doi.org/10.1016/j.asej.2024.102725).
- [3] W. Zeng, K. Wang, J. Zhou, R. Cheng, Traffic flow prediction based on hybrid deep learning models considering missing data and multiple factors, *Sustainability* 15 (14) (2023) 11092. [doi:10.3390/su151411092](https://doi.org/10.3390/su151411092).
- [4] I. Tunc, M. T. Soylemez, Fuzzy logic and deep q-learning based control for traffic lights, *Alexandria Engineering Journal* 67 (2023) 343–359. [doi:10.1016/j.aej.2022.12.028](https://doi.org/10.1016/j.aej.2022.12.028).
- [5] W. Wang, J. Shao, H. Jumahong, Fuzzy inference-based lstm for long-term time series prediction, *Scientific Reports* 13 (1) (2023) 20359. [doi:10.1038/s41598-023-47812-3](https://doi.org/10.1038/s41598-023-47812-3).

- [6] S. M. Abdullah, M. Periyasamy, N. A. Kamaludeen, S. K. Towfek, R. Marappan, S. Kidambi Raju, A. H. Alharbi, D. S. Khafaga, Optimizing traffic flow in smart cities: Soft gru-based recurrent neural networks for enhanced congestion prediction, *Sustainability* 15 (7) (2023) 5949. doi:10.3390/su15075949.
- [7] C. R. Goenawan, ASTM: Autonomous smart traffic management system using AI CNN and LSTM, arXiv preprint arXiv:2401.10929 (2024). doi:10.48550/arXiv.2401.10929.
- [8] A. Kadkhodayi, M. Jabeli, H. Aghdam, S. Mirbakhsh, Artificial intelligence-based real-time traffic management, *Journal of Electrical and Electronic Engineering* 2 (4) (2023) 368–373. URL <https://www.researchgate.net/publication/375006146>
- [9] H.-Y. Lin, B.-W. Hsu, Application of hybrid fuzzy interval-based machine learning models on financial time series, *Frontiers in Artificial Intelligence* 6 (2024) 1283741. doi:10.3389/frai.2023.1283741.
- [10] V. Gandrybida, D. Bondarenko, V. Sevastyanov, Advancements in automated traffic management using fuzzy logic: prospects and challenges, *Information Technologies and Computer Engineering* 3 (21) (2024) 86–95. URL https://itce.com.ua/web/uploads/pdf/%D0%86%D0%A2%D0%9A%D0%86_21_3_2024_newDOI_86-95.pdf
- [11] A. N. Abood, Traffic light control in vehicular network systems using fuzzy logic, *European Journal of Information Technologies and Computer Science* 4 (4) (2024) 1–6. doi:10.24018/compute.2024.4.4.132.
- [12] Preetham Gouda, Bangalore's Traffic Dataset, accessed: 2024-09-21 (2023). URL <https://www.kaggle.com/datasets/preethamgouda/bangalore-city-traffic-dataset>
- [13] H. Chen, Z. Lin, Y. Yao, H. Xie, Y. Song, T. Zhou, Hybrid extreme learning for reliable short-term traffic flow forecasting, *Mathematics* 12 (20) (2024) 3303. doi:10.3390/math12203303.
- [14] A. R. Sattarzadeh, R. J. Kutadinata, P. N. Pathirana, V. T. Huynh, A novel hybrid deep learning model with arima, conv-lstm networks and shuffle-attention layer for short-term traffic flow prediction, *Transportmetrica A: Transport Science* 21 (1) (2023) 2236724. doi:10.1080/23249935.2023.2236724.